

Introduction to R

Research Methods and Skills

06/10/2026

Who are we?

Dr Tochukwu Onwuegbusi



Sarah Swift Building Room 4227
tonwuegbusi@lincoln.ac.uk

Luke Galloway



Sarah Swift Building Room 4225
lgalloway@lincoln.ac.uk

Group A (PSY9251M) Skills Audit

We would like to invite you to complete the Research Method Skill Audit? This should take no longer than 4 minutes. The audit is important as it helps us to tailor our teaching and support to meet your learning needs.

All answers given are anonymous and are used to reflect on your statistical and programming skills, and for us to gauge what skills shortage need addressing so that we can help you to the best of our ability. Please use the link below to complete the short audit questionnaire!

<https://forms.cloud.microsoft/e/cZKwUPSfVm>

Group B (PSY9219M) Skills Audit

We would like to invite you to complete the Research Method Skill Audit? This should take no longer than 4 minutes. The audit is important as it helps us to tailor our teaching and support to meet your learning needs.

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<https://forms.cloud.microsoft/e/cZKwUPSfVm>

Scenario A

You've just started work in a psychology lab. You're asked to help analyse some old data. There is reaction time data from 50 participants. Each participant's data is stored in a separate text file.

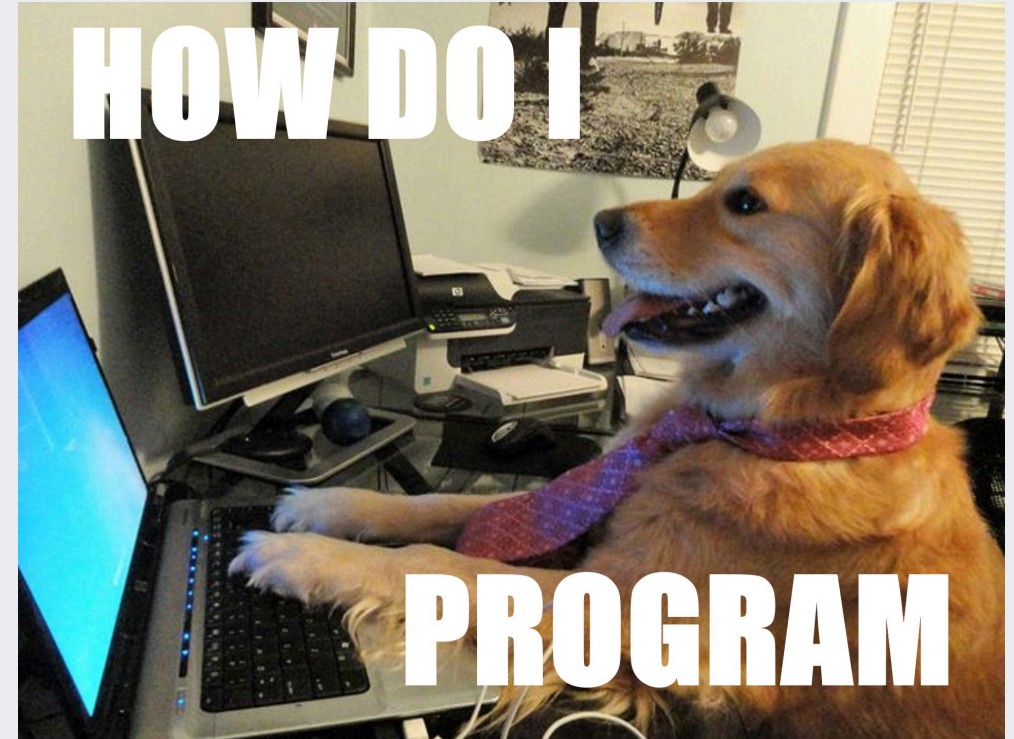
- How do you combine the data from each participant together to be able to analyse the data?
- It turns out some of the participants only completed part of the experiment - which ones, and what should you do with their data?
- What steps should you take to select and perform appropriate statistical analysis?

Scenario B

You've been asked to design, implement, and evaluate a new treatment regime across several psychiatric institutes. Several colleagues are skeptical that it can deliver the kind of improvements in outcomes indicated in a publication describing the method.

- How do you interpret the strength of the previously published evidence?
- How do you design a rigorous test of the treatment efficacy?
- How do you evaluate and report on the outcomes of your trial?

Research Methods and Skills



Course outline

Weeks 2-7

- Introduction to R
- Basic R programming
- Plotting with ggplot2
- Data import, selection and manipulation
- Describing and summarising your data

Weeks 12-16

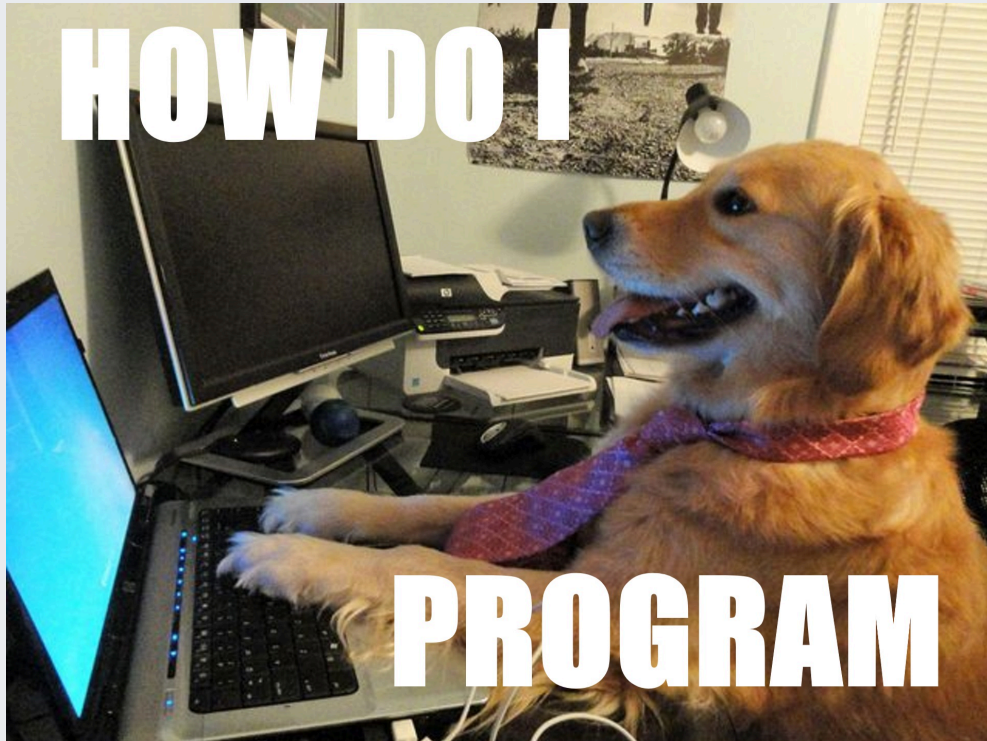
- Qualitative methods

Weeks 8-11

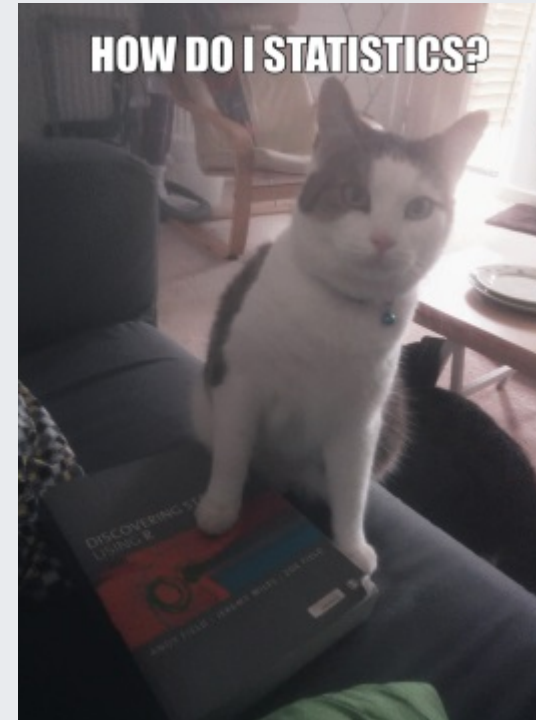
- Hypothesis testing and estimation
- *t*-tests and comparing two groups
- Correlation and linear regression
- One-way ANOVA
- Factorial ANOVA

Course outline

Weeks 2-7

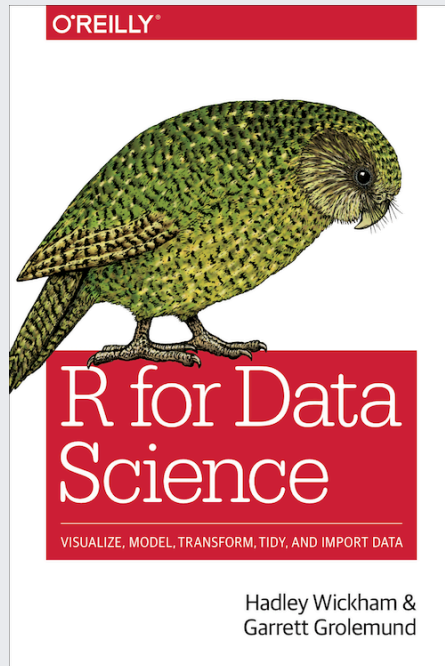


Weeks 7, 8-10

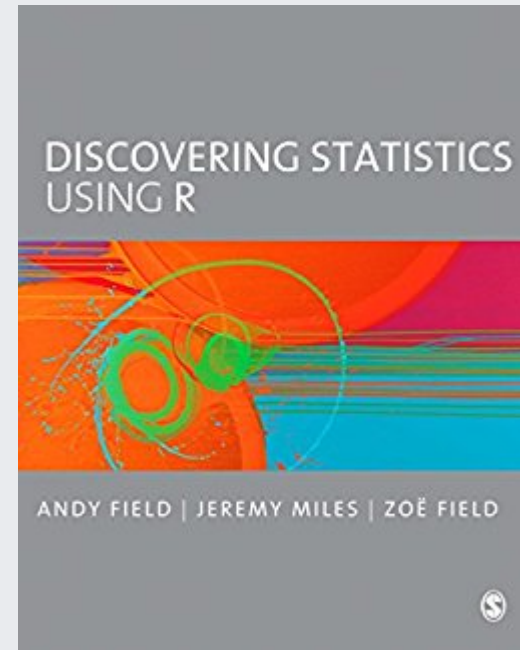


Course outline

Weeks 2-7



Weeks 8-11



Semester B - Advanced Research Methods and Skills (Course outline)

Weeks 19-23

- Module Intro/Approaches to knowledge
- Generalisability and Open Science
- Systematic Literature Review
- Qual 1: Reflexivity/Intersectionality/Thematic analysis
- Qual 2: Thematic/Narrative analysis

Weeks 24-29

- ANOVA/Linear Regression
- Logistic Regression
- Mixed effects regression
- Factor Analysis
- Handling large & messy datasets
- Visualisation/Review/Assessment Q&A

Weeks 30-33

- Interpreting Results/Review/Assessment Q&A
- Optional drop-in

How it's going to work

Where possible you'll be provided with a set of pre-recorded videos each week introducing the core topics for that week.

Most weeks you will be provided with some exercises to work on and complete formative assignment. The aim of the formative stage of the assignment is to enable you to produce an outline of your proposed summative for feedback from the module team.

In the timetabled sessions, we will spend the first hour (lecture) going over the topic for that week. In the second hour (workshop), you can spend the time working together on the exercises, and we'll be available to help out. You will finish up with a formative assignment for that week and receive a formative feedback before the next teaching session.

If you get stuck **ask us!**

Introduction to R and RStudio

What is R?

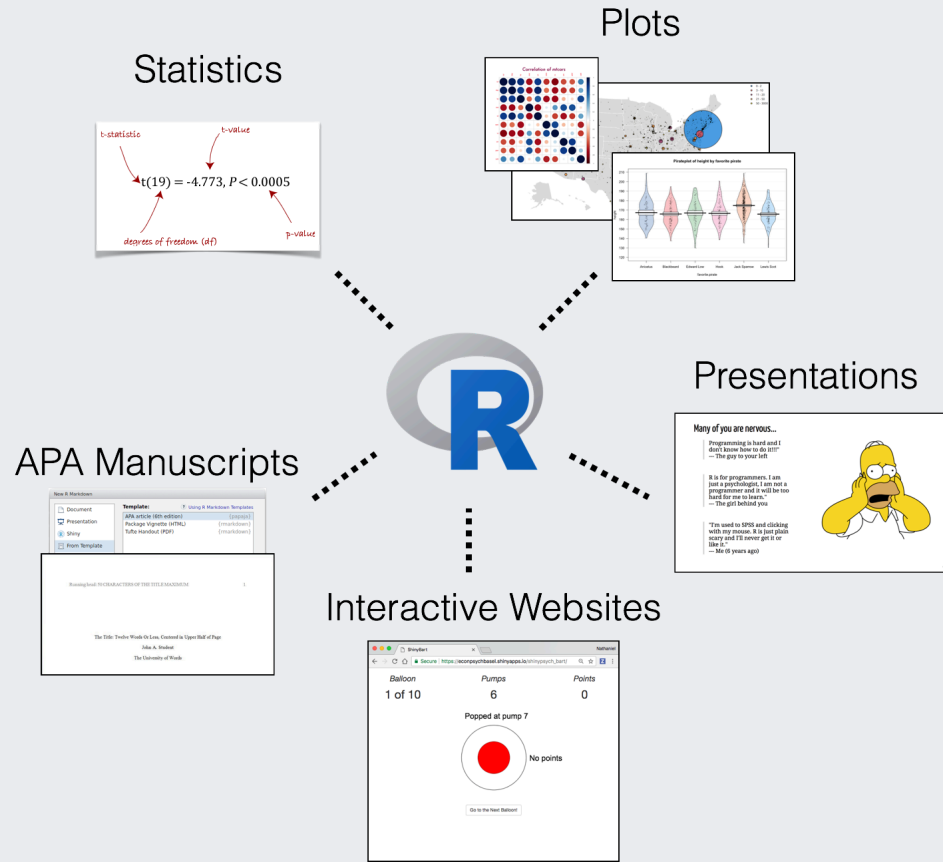


R is a statistical, mathematical programming language

- Created in 1993
- Designed from the ground up to support many statistical tasks
- Covers all aspects of data analysis from import through to production of reports
- Free, open source
 - Can be downloaded from the R-project website
- Continually evolving
 - R has over 25,000 *packages* that add additional functions

But WHY?

What can you do with R?

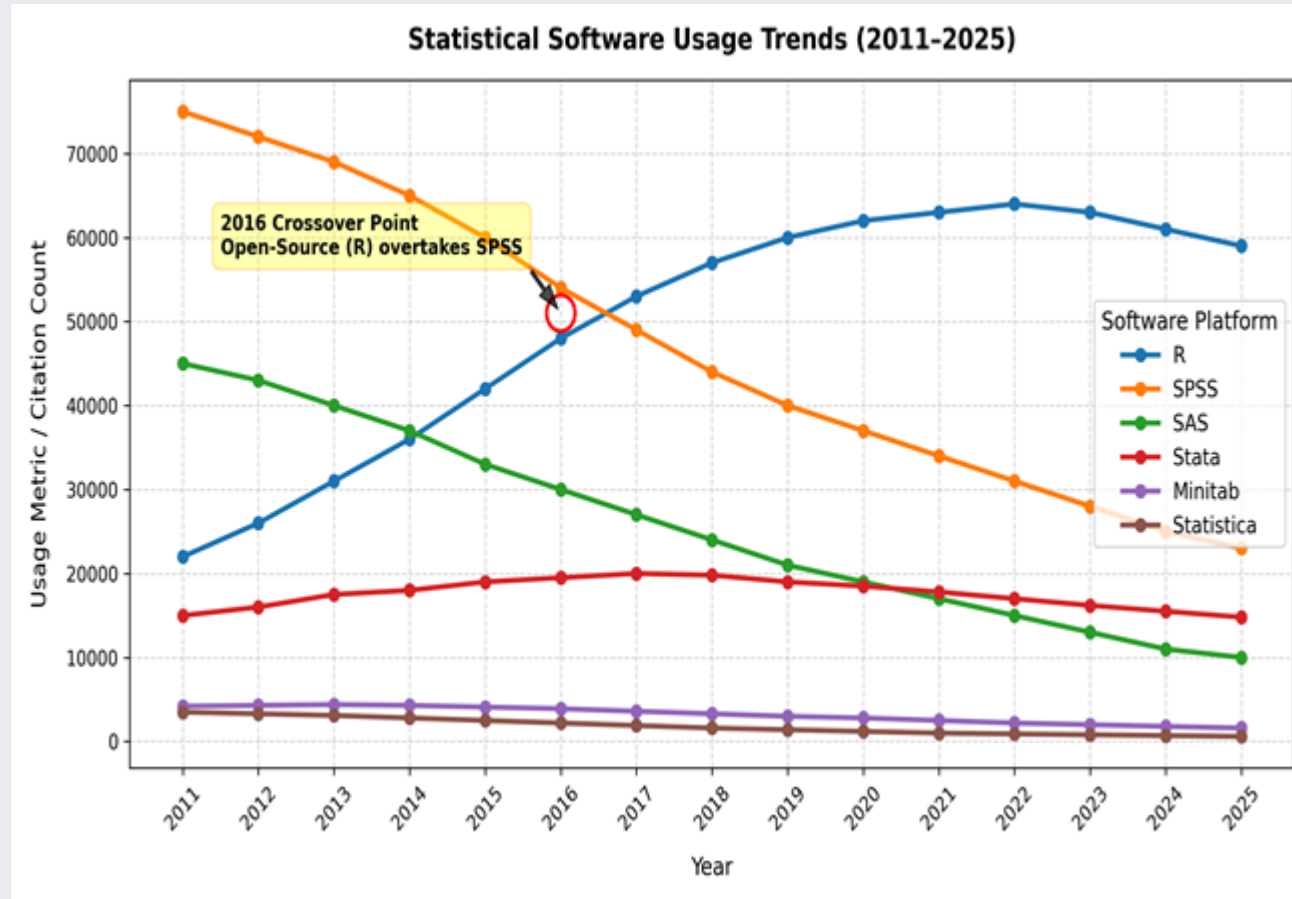


Source: N. D. Phillips

Companies that use R for Analytics



Still not convinced?

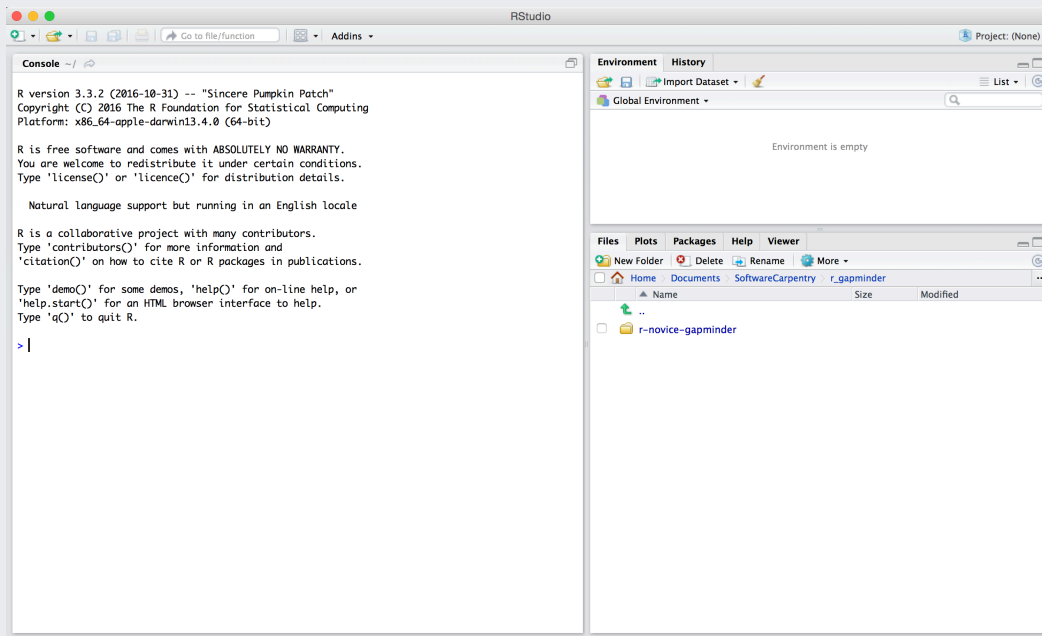


Getting started

What is RStudio?

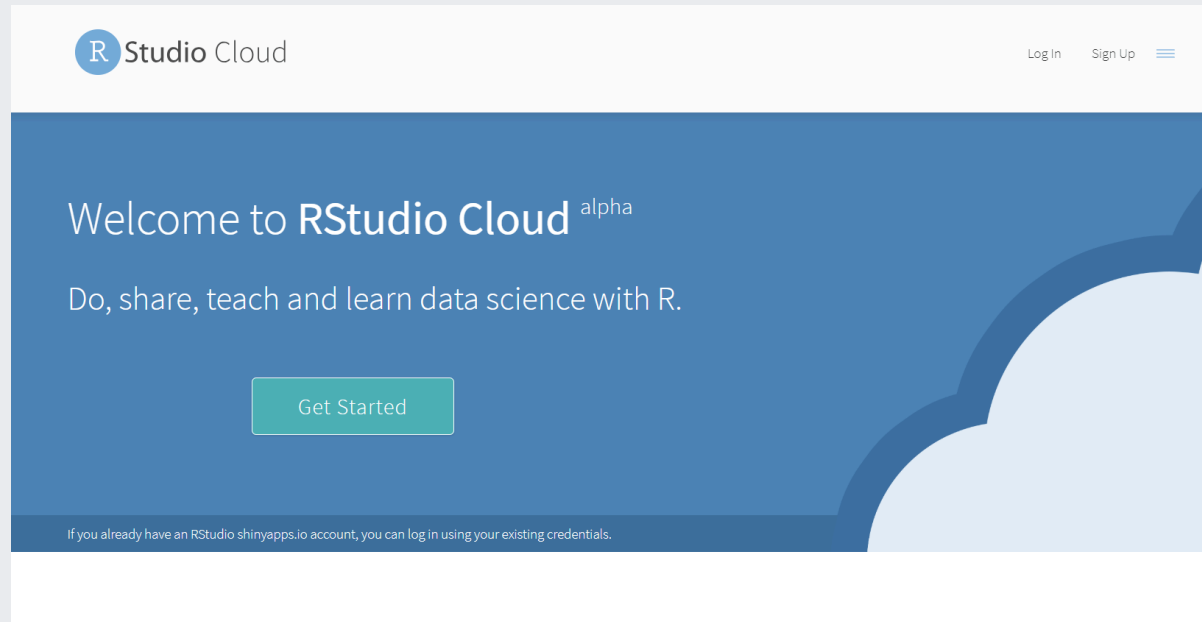


- An **Integrated Development Environment (IDE)**
- An interface for R that makes your life much, much easier
- Makes many things explicit that you would otherwise have to guess



Getting started TODAY

1. Open up a web browser
2. Go to <https://posit.cloud/>
3. Sign up! Use your REAL NAME, and your University of Lincoln email address.





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   Go to file/function  Addins

R 3.5.0

Console Terminal x Jobs x

/cloud/project/

```
R version 3.5.0 (2018-04-23) -- "Joy in Playing"
Copyright (C) 2018 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
```

```
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Type 'license()' or 'licence()' for distribution details.
```

```
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
```

```
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
```

>

Environment History Connections

 Import Dataset

List

Global Environment

Environment is empty

Files Plots Packages Help Viewer

 New Folder  Upload  Delete  Rename  More

Cloud > project

	Name	Size	Modified
	..		
	.Rhistory	0 B	Sep 18, 2018
	project.Rproj	205 B	Sep 18, 2018

ConsoleTerminal xJobs x

/cloud/project/

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> |

Install Packages...

Check for Package Updates...

Version Control

Shell...

Terminal

Addins

Keyboard Shortcuts Help Shift+Alt+K

Modify Keyboard Shortcuts...

Project Options...

Global Options...

Connections

Dataset

Environment is empty

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RStudio Cloud

https://rstudio.cloud/project/84901

Your Workspace / Untitled Project

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Addins

R 3.5.0

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'citation()' on how to cite R or

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'help.start()' for an HTML brows
Type 'q()' to quit R.

> |

Options

GeneralCodeAppearancePane LayoutPackagesR MarkdownSweaveSpellingGit/SVNPublishingTerminal

RStudio theme:Modern

Editor Font size:10

Editor theme:
Ambiance
Chaos
Chrome
Clouds Midnight
Clouds
Cobalt
Crimson Editor
Dawn
Dracula
Dreamweaver
Eclipse
Idle Fingers
Katzenmilch
Kr Theme
Material

```
# plotting of R objects
plot <- function(x, y, ...)
{
  if (is.function(x) &&
      is.null(attr(x, "class")))
  {
    if (missing(y))
      y <- NULL

    # check for ylab argument
    hasylab <- function(...)
      !all(is.na(
        pmatch(names(list(...)),
                  "ylab")))

    if (hasylab(...))
      plot.function(x, y, ...)

    else
      plot.function(
        x, y,
        ylab = paste(
          deparse(substitute(x)),
          "(x)",
          ...)
      )
  }
  else
    UseMethod("plot")
}
```

OKCancelApply

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er

RenameMore

Size	Modified
0 B	Sep 24, 2018, 11:53 AM
205 B	Sep 24, 2018, 12:29 PM

How it works!

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```

>

Environment History Connections



Import Dataset



List



Global Environment



Environment is empty

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More



Cloud > project

	Name	Size	Modified
	..		
☐	.Rhistory	0 B	Sep 18, 2018
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+ Go to file/function Addins

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Console Terminal x Jobs x

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	..		
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	project.Rproj	205 B	Sep 18, 2018

REPL

How to use R

- The R Console
 - REPL: Read/Evaluate/Print/Loop
 - Type stuff in, it tries to do it

When you see the > symbol -

```
>
```

... R is waiting for your input.

```
> 5  
[1] 5
```

Warming up

Try using R like a calculator!

Basic arithmetic operators

Symbol	Operation
+	addition
-	subtraction
*	multiplication
/	division
^	exponentiation
%%	modulo

Warming up

You can break up long maths expressions over multiple lines:

```
2 + 4 + 5 +  
  5 + 6 + 7 + 8 +  
 10
```

```
## [1] 47
```

Note that when you do that, the ">" symbol changes to a "+"

```
> 5 +  
+ 5  
[1] 10
```

Remember!

> means R is waiting for input.

```
>
```

+ means R is waiting for you to finish your command.

```
+
```

Either finish your command, or press the **Esc** key to cancel it.

Text input

R can also accept text strings as input.

```
"hello world!"  
[1] "hello world!"
```

You need to use quotation marks (") to tell R that this is text:

```
hello world!
```

```
## Error: <text>:1:7: unexpected symbol  
## 1: hello world  
##           ^
```

Otherwise, you'll receive an error like the one above.

The assignment operator

In R, you can assign values to an **object** for subsequent use. **Objects** have names that are written as text.

The assignment operator is the two-character symbol:

```
<-
```

You assign values to objects by putting the `<-` sign between the name of the object and the value you want to give it:

```
example <- 5  
example
```

```
## [1] 5
```

Note that R does not immediately provide output when you assign the output to an object.

The assignment operator

Think of `<-` as meaning "is now". i.e.

```
example <- 5
```

can be read as

```
The object "example" is now 5
```

Working with objects

Once an object is assigned, the name that you gave it *stands in* for the *value* that you assigned to it, and can be used as if it were that value:

```
example
```

```
## [1] 5
```

```
example + 10
```

```
## [1] 15
```

```
example + 13 - 1 * 2 %% 4
```

```
## [1] 16
```

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Type 'q()' to quit R.

>
> example <- 10
> hi_there <- "hi there!"
> hi_there
[1] "hi there!"
> example
[1] 10
> |

EnvironmentHistoryConnections

Global Environment

Values

example	10
hi_there	"hi there!"

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	Name	Size	Modified
	..		
☐	.Rhistory	0 B	Sep 24, 2018, 11:53 AM
☐	project.Rproj	205 B	Sep 24, 2018, 1:06 PM

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```
>  
> example <- 10  
> hi_there <- "hi there!"  
> hi_there  
[1] "hi there!"  
> example  
[1] 10  
> |
```

Environment History Connections

Global Environment

Values

example	10
hi_there	"hi there!"

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Cloud > project

	Name	Size	Modified
	..		
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```
>  
> example <- 10  
> hi_there <- "hi there!"  
> hi_there  
[1] "hi there!"  
> example  
[1] 10  
>
```

Environment History Connections

example <- 10
hi_there <- "hi there!"
hi_there
example

Files Plots Packages Help Viewer

.Rhistory 0 B Sep 24, 2
project.Rproj 205 B Sep 24, 2

History

Try it out!

Let's try a few things out!

1. Assign some values to objects using the assignment operator (<-)
2. Try using arithmetic operations (e.g. *, /, %%) on those objects
3. Try using arithmetic operations to combine multiple numerical objects
4. Try using arithmetic operations on text

Combining multiple things

Sometimes you want to allocate more than one value to an object. You can use the `c()` function to do this.

```
c(8, 5, 10)
```

```
## [1] 8 5 10
```

```
example <- c(8, 5, 10)  
example
```

```
## [1] 8 5 10
```

```
c("hello", "how", "are", "you")
```

```
## [1] "hello" "how" "are" "you"
```

IMPORTANT: BRACKETS () AFTER A WORD MEAN THAT THIS IS A FUNCTION

Vectors

The function `c()` is creating **vectors**.

Vectors are simply a one-dimensional collection of things that all have the same *type* (we will cover data types next week!).

Note that mixing, for example, text and numbers, will yield a *character* vector.

```
c(5, "five", 2)
```

```
## [1] "5"      "five" "2"
```

Functions

Functions are commands that operate on **objects**.

For example, to calculate the *mean* of several numbers, you can use the function **mean()**. The output of functions can also be assigned to **objects** using **<-**.

```
mean(c(8, 5, 10))
```

```
## [1] 7.666667
```

```
example <- c(8, 5, 10)  
mean(example)
```

```
## [1] 7.666667
```

```
example_mean <- mean(example)  
example_mean
```

```
## [1] 7.666667
```

Let's try it out!

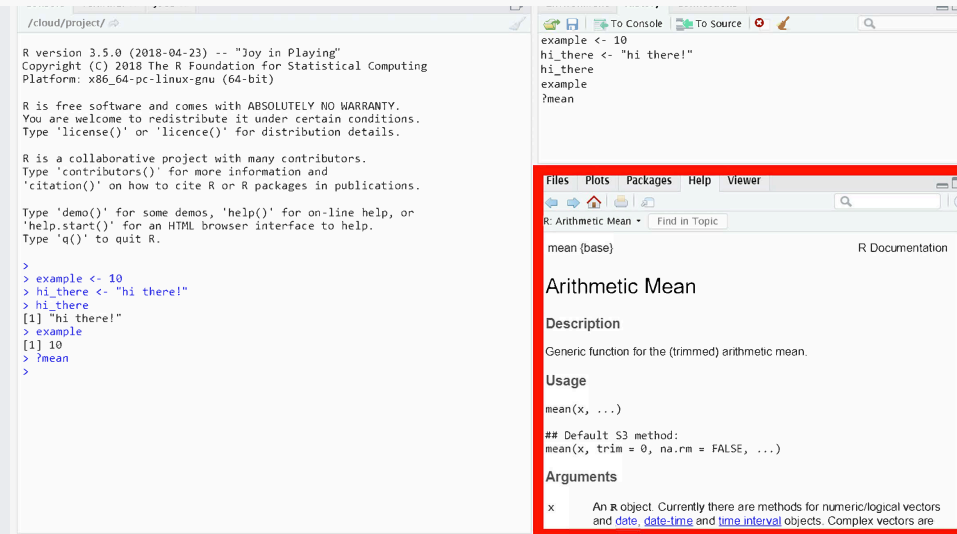
1. Use **c()** to create a vector of numbers.
2. Use **c()** to create a vector of strings.
3. Calculate the **mean()** of a vector of numbers.
4. Try guessing some other simple statistics (e.g. other types of *average*) that you can use.

Getting help

If you don't know how to use a function, R has built-in help!

There are several ways you can access it:

```
help("mean")  
?mean  
??mean
```



Packages

Packages are the key to R's versatility. Over 14000 are currently available from the **Comprehensive R Archive Network** - CRAN. The `install.packages()` function can be used to install packages.

Let's install the "cowsay" package. **cowsay** is an extraordinarily useful package, as you'll see.

One way to install the package is using the console:

```
install.packages("cowsay")
```

Once it's installed, use the `library()` function to load the package!

```
library(cowsay)
```

But **another** way to install is using the GUI!

Still on Packages



Top Level (Base R / CEO): Base R acts as the Chief Executive Officer overseeing the entire operation. It provides the core foundation and environment where everything runs.

Second Layer (Departmental Managers/R Packages): The second layer shows the specialized managers: Ian (Marketing), and Dave (Finance). In R, these are your specialized packages—like `readxl` and `ggplot2`. Just as you wouldn't ask Ian in Marketing to handle salary queries, you wouldn't use `readxl` to build a bar chart.

Third Layer (Responsibilities & Specific Functions): The third layer shows the specific responsibilities for each department. In R, these are your functions: To import Excel file, you load 'readxl' package and use 'read_excel()'

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> example <- 10
> hi_there <- "hi there!"
> hi_there
[1] "hi there!"
> example
[1] 10
> ?mean
> |

Environment History Connections

example <- 10
hi_there <- "hi there!"
hi_there
example
?mean

Files Plots Packages Help Viewer

Install Update Packrat

Name	Description	Version
System Library		
☐ boot	Bootstrap Functions (Originally by Angelo Canty for S)	1.3-20
☐ class	Functions for Classification	7.3-14
☐ cluster	"Finding Groups in Data": Cluster Analysis Extended Rousseeuw et al.	2.0.7-1
☐ codetools	Code Analysis Tools for R	0.2-15
☐ compiler	The R Compiler Package	3.5.0
☒ datasets	The R Datasets Package	3.5.0
☐ foreign	Read Data Stored by 'Minitab', 'S', 'SAS', 'SPSS', 'Stata', 'Systat', 'Weka', 'dBase', ...	0.8-70
☒ graphics	The R Graphics Package	3.5.0
☒ grDevices	The R Graphics Devices and Support for Colours and Fonts	3.5.0
☐ grid	The Grid Graphics Package	3.5.0
☐ KernSmooth	Functions for Kernel Smoothing Supporting Wand & Jones (1995)	2.23-15
☐ lattice	Trellis Graphics for R	0.20-35
☐ MASS	Support Functions and Datasets for Venables and Ripley's MASS	7.3-49

```
example <- 10
hi_there <- "hi there!"
hi_there
example
?mean
```

Install Packages

Install from: [? Configuring Repositories](#)

Repository (CRAN, RSPM) ▼

Packages (separate multiple with space or comma):

cowsay

`cowsay` Library:

/home/rstudio-user/R/x86_64-pc-linux-gnu-library/3.5 [Default] ▼

☒ Install dependencies

Install

Cancel

☒ datasets☐ foreign☒ graphics☒ grDevices☐ grid☐ Kern☐ lattice

MASS

es Help Viewer

 Packrat

Item	Description
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Bootstrap Functions (Originally by Angelo Canty for S)

Functions for Classification	7.3-14
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"Finding Groups in Data": Cluster Analysis 2.0.7-1

Extended Rousseeuw et al.

Code Analysis Tools for R 0.2-15

The R Compiler Package 3.5.0

The R Datasets Package	3.5.0
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Read Data Stored by 'Minitab', 'S', 'SAS',	0.8-70
--	--------

'SPSS', 'Stata', 'Systat', 'Weka', 'dBase', ...

The R Graphics Package 3.5.0

The R Graphics Devices and Support for 3.5.0

Colours and Fonts

The Grid Graphics Package 3.5.0

Functions for Kernel Smoothing 2.23-15

Supporting wand & Jones (1995)	
Trellis Graphics for R	0.30.25

Trends Graphics for R 0.20-35	0.20-35
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Support Functions and Datasets for Variables and Binsley's MASS 7.3-49

Let's try out the cowsay package

cowsay adds a function called **say()**. Load the package in as follows, and look at the help for **say()**.

```
## Warning: package 'cowsay' was built under R version 4.2.2
```

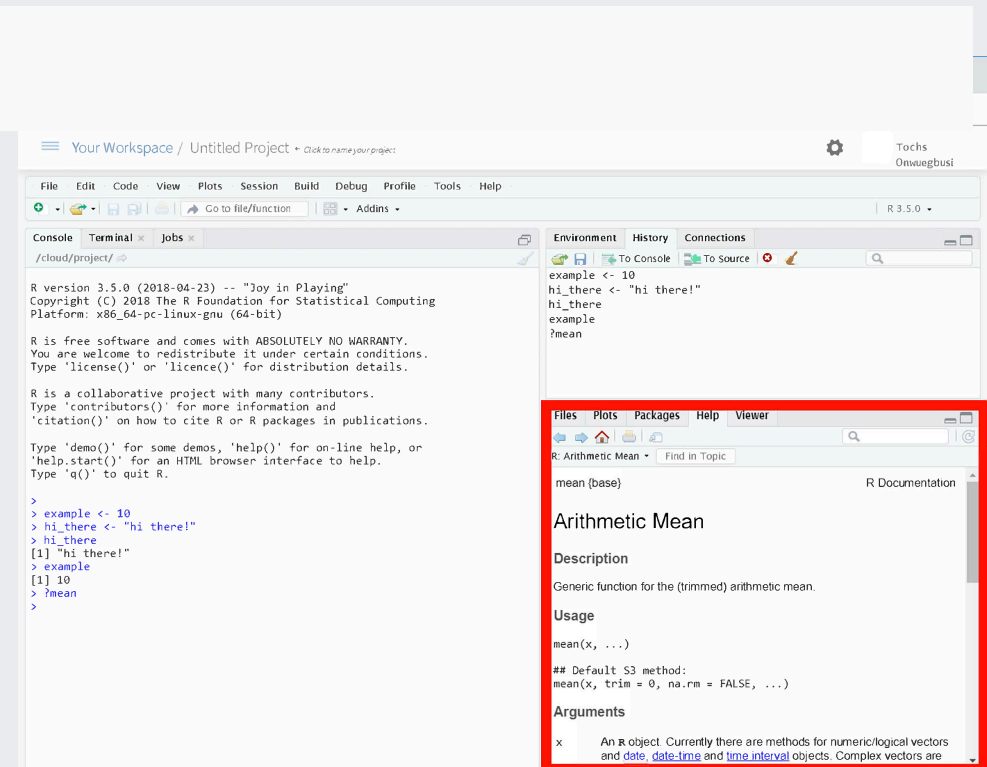
```
library(cowsay)
?say()
```

Remember that help appears in the bottom right window!

Look at **Usage** and **Arguments**

Usage is how to use the function.

Arguments are what the functions expect and understand.



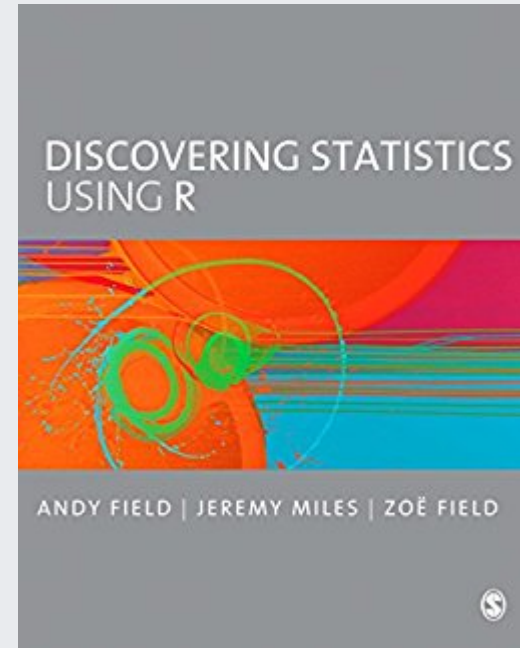
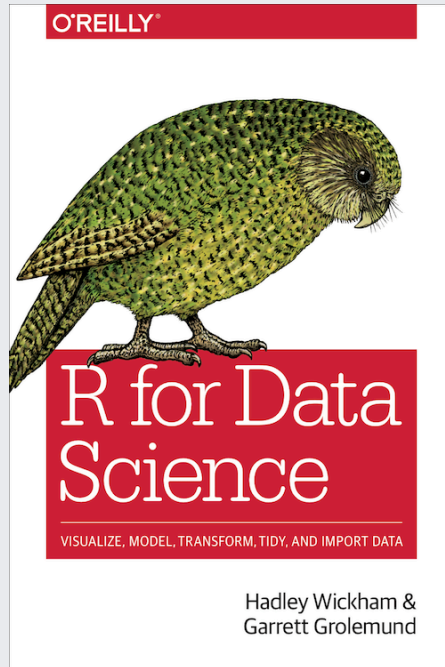
```
say(what = "Feed me, human.", by = "cat")
```

```
##
## -----
## Feed me, human.
## -----
##      \
##      \
##      \
##      | \_ _ _ / |
##      ==) ^Y^ (==
##          ^
##          \   /
##          )=*= (
##          /     \
##          |       |
##          /| | | | \
##          \| | | | /
##      jgs // _ // _ _ _ /
##          \_ )
##
```

Try out the say() function

1. Try a few different animals by changing the **by** argument
2. Change what the animals say by changing the **what** argument.

Additional resources



There are copies of both these books in the library.

R for Data Science is available freely online at <http://r4ds.had.co.nz/>

An additional recommendation...

Hands-on Programming with R

Basic R programming book, also available for free online

<https://rstudio-education.github.io/hopr/index.html>



This week's goals

1. Download R and RStudio! You'll find links and instructions on Blackboard.
2. Read through Chapter 1 of R for Data Science
3. Try out some of the introductory things mentioned in the slides, just to get a feel for using RStudio!